

chap_grappe_SIC_effetGrappe

2025-07-15

Mise en oeuvre SIC

```
library(sampling)
data(rec99)
(NI<-length(table(rec99$BVQ_N)))

## [1] 32
range(table(rec99$BVQ_N))

## [1] 5 63
#[1] 5 63

nI <- 4
set.seed(123456)
grap.rec <- sampling::cluster(rec99,clustername=c("BVQ_N"),
                             size=nI,method="srswor",description=TRUE)

## Number of selected clusters: 4
## Number of units in the population and number of selected units: 554 79
#Number of selected clusters: 4
#Population total and number of selected units 554 39

grap.data <- getdata(rec99, grap.rec)
range(grap.data$Prob)

## [1] 0.125 0.125
#[1] 0.125 0.125

table(grap.data$BVQ_N)

##
## 31113 31358 31555 31582
##    12    13    14    40
#31113 31144 31471 31499
#12 6 10 11

library(survey)

## Loading required package: grid
## Loading required package: Matrix
## Loading required package: survival
##
## Attaching package: 'survival'
```

```
## The following objects are masked from 'package:sampling':
##
##   cluster, strata
##
## Attaching package: 'survey'
##
## The following object is masked from 'package:graphics':
##
##   dotchart
ech.grap <- svydesign(ids=~BVQ_N, probs = ~Prob, data=grap.data,
  fpc=~rep(nI/NI, nrow(grap.data)))
(est.grappe<-svytotal(~LOGVAC, ech.grap))

##           total      SE
## LOGVAC 13472 2358
#           total      SE
#LOGVAC 7368 1815.8
SE(est.grappe)/coef(est.grappe)

##           LOGVAC
## LOGVAC 0.1750291
#           LOGVAC
#LOGVAC 0.2464406

aggregate(grap.data[,4:5],by=list(grap.data$BVQ_N),sum)

##   Group.1  LOG LOGVAC
## 1  31113  6834   302
## 2  31358  5288   271
## 3  31555 17389   584
## 4  31582  9174   527
#   Group.1  LOG LOGVAC
#1  31113  6834   302
#2  31144  927    72
#3  31471  2465   202
#4  31499  8857   345
```

Effet de grappe et efficacité de la stratégie (SIC,HT)

```
rec99$BVQ_N <- as.factor(rec99$BVQ_N)
summary(lm(rec99$LOGVAC~rec99$BVQ_N,data=rec99))

##
## Call:
## lm(formula = rec99$LOGVAC ~ rec99$BVQ_N, data = rec99)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -39.17  -13.86   -6.36    0.73   335.00
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      16.300     10.307   1.582  0.1144
```

```

## rec99$BVQ_N31028 -7.888 12.989 -0.607 0.5439
## rec99$BVQ_N31033 5.657 12.345 0.458 0.6470
## rec99$BVQ_N31042 -1.300 11.725 -0.111 0.9118
## rec99$BVQ_N31044 -1.729 16.062 -0.108 0.9143
## rec99$BVQ_N31080 -9.027 12.430 -0.726 0.4680
## rec99$BVQ_N31098 -7.514 13.494 -0.557 0.5779
## rec99$BVQ_N31106 -5.031 12.128 -0.415 0.6784
## rec99$BVQ_N31107 10.867 12.855 0.845 0.3983
## rec99$BVQ_N31113 8.867 13.955 0.635 0.5255
## rec99$BVQ_N31135 9.367 12.522 0.748 0.4548
## rec99$BVQ_N31144 -4.300 16.831 -0.255 0.7984
## rec99$BVQ_N31193 -10.069 13.709 -0.734 0.4630
## rec99$BVQ_N31202 8.575 15.460 0.555 0.5794
## rec99$BVQ_N31232 23.874 12.345 1.934 0.0537 .
## rec99$BVQ_N31239 -6.133 12.267 -0.500 0.6173
## rec99$BVQ_N31358 4.546 13.709 0.332 0.7403
## rec99$BVQ_N31375 11.575 15.460 0.749 0.4544
## rec99$BVQ_N31381 -8.900 17.851 -0.499 0.6183
## rec99$BVQ_N31390 16.300 13.306 1.225 0.2211
## rec99$BVQ_N31395 20.613 12.345 1.670 0.0956 .
## rec99$BVQ_N31451 14.245 14.241 1.000 0.3176
## rec99$BVQ_N31454 -2.113 13.138 -0.161 0.8723
## rec99$BVQ_N31471 3.900 14.576 0.268 0.7891
## rec99$BVQ_N31483 -4.538 11.094 -0.409 0.6827
## rec99$BVQ_N31499 15.064 14.241 1.058 0.2906
## rec99$BVQ_N31523 3.780 12.195 0.310 0.7567
## rec99$BVQ_N31555 25.414 13.494 1.883 0.0602 .
## rec99$BVQ_N31561 19.256 14.975 1.286 0.1991
## rec99$BVQ_N31573 -3.133 16.831 -0.186 0.8524
## rec99$BVQ_N31582 -3.125 11.523 -0.271 0.7863
## rec99$BVQ_N31584 10.414 16.062 0.648 0.5170
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.59 on 522 degrees of freedom
## Multiple R-squared: 0.09216, Adjusted R-squared: 0.03825
## F-statistic: 1.709 on 31 and 522 DF, p-value: 0.01084
#
#---
#Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
#Residual standard error: 32.59 on 522 degrees of freedom
#Multiple R-squared: 0.09216, Adjusted R-squared: 0.03825
#F-statistic: 1.709 on 31 and 522 DF, p-value: 0.01084

summary(aov(rec99$LOGVAC~rec99$BVQ_N,data=rec99))

## Df Sum Sq Mean Sq F value Pr(>F)
## rec99$BVQ_N 31 56292 1816 1.709 0.0108 *
## Residuals 522 554496 1062
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
#Df Sum Sq Mean Sq F value Pr(>F)
#rec99$BVQ_N 31 56292 1816 1.709 0.0108 *
#Residuals 522 554496 1062
#---
#Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
SSB <- 56292; SSW <- 554496; SST <- SSB+SSW
SyU2 <- SST / (554 - 1) ; SyW2 <- SSW / (554 - 32)
(delta <- 1 - SyW2 / SyU2)
```

```
## [1] 0.03824921
```

```
#[1] 0.03824921
```

Plan en grappes de taille fixe sans remise et proportionnel à la taille

Plan STSI de grappes

```
o <- as.data.frame(table(rec99$BVQ_N))
names(o) <- c("BVQ_N", "Ni")
rec_st <- merge(rec99, o)
rec_st$stratBVQ <- ifelse(rec_st$Ni < 20, 1, 2)
rec_st[1:3,]
```

```
##   BVQ_N CODE_N      COMMUNE POPSDC99 LOG LOGVAC STRATLOG surf_m2 lat_centre
## 1 31020 31140 CHEIN-DESSUS      184 174      28        2 12743364 43.00497
## 2 31020 31245 JUZET-D'IZAU      193 153        6        2 14884960 42.96997
## 3 31020 31020      ASPET       923 684      90        3 26525625 43.01137
##   lon_centre      NAME_3 Ni stratBVQ
## 1 0.8752188 Saint-Gaudens 10        1
## 2 0.7575574 Saint-Gaudens 10        1
## 3 0.8062536 Saint-Gaudens 10        1
```

```
#BVQ_N CODE_N COMMUNE POPSDC99 LOG LOGVAC stratlog Ni stratBVQ
#1 31020 31014 ARGUENOS 57 94 1 1 10 1
#2 31020 31131 CAZAUNOUS 47 56 4 1 10 1
#3 31020 31348 MONCAUP 26 57 2 1 10 1
```

```
NBVQ <- table(rec_st$stratBVQ, rec_st$BVQ_N)
NBVQ <- ifelse(NBVQ==0, 0, 1)
(NBVQ <- apply(NBVQ, 1, sum))
```

```
## 1 2
```

```
## 21 11
```

```
#1 2
```

```
#21 11
```

```
n.grappe <- 2
set.seed(123456)
grap.rec1 <- sampling::cluster(rec_st[rec_st$stratBVQ==1,],
  clustername=c("BVQ_N"), size=n.grappe,
  method="srswor", description=TRUE)
```

```
## Number of selected clusters: 2
```

```
## Number of units in the population and number of selected units: 230 20
```

```

#Number of selected clusters: 2
#Number of units in the population and number of selected units:
#230 21
grap2.rec1 <- getdata(rec_st[rec_st$stratBVQ==1,], grap.rec1)
(nrow(grap2.rec1))

## [1] 20

#[1] 21

grap.rec2 <- sampling::cluster(rec_st[rec_st$stratBVQ==2,],
                               clustername=c("BVQ_N"),size=n.grappe,
                               method="srswor",description=TRUE)

## Number of selected clusters: 2
## Number of units in the population and number of selected units: 324 47

#Number of selected clusters: 2
#Number of units in the population and number of selected units:
#324 47
grap2.rec2 <- getdata(rec_st[rec_st$stratBVQ==2,], grap.rec2)
(nrow(grap2.rec2))

## [1] 47

#[1] 47

grapSTSI.rec <- rbind(grap2.rec1,grap2.rec2)

(pik.strate<-n.grappe/NBVQ)

##          1          2
## 0.0952381 0.1818182

#1 2
#0.0952381 0.1818182

ech.grapSTSI <- svydesign(ids=~BVQ_N,strata =~stratBVQ,
                          probs=~Prob, data=grapSTSI.rec,
                          fpc=~c(rep(pik.strate[1],nrow(grap2.rec1)),
                                rep(pik.strate[2],nrow(grap2.rec2))))

(est.LOGVAC<-svytotal(~LOGVAC, ech.grapSTSI))

##          total      SE
## LOGVAC 8928.5 1545.9

#          total      SE
#LOGVAC 10320 1880.8
SE(est.LOGVAC)/coef(est.LOGVAC)

##          LOGVAC
## LOGVAC 0.1731431

#          LOGVAC
#LOGVAC 0.1822608

```

Plan en grappes proportionnel à la taille

```
data(swissmunicipalities)
sum(swissmunicipalities$Surfacesbois)

## [1] 1270996
# [1] 1270996

swiss1 <- swissmunicipalities[,c(1,5,6)]
swiss1[1:5,]

##   CT HApoly Surfacesbois
## 1  1   8781         2326
## 2 25   1593          67
## 3 12   2391          97
## 4  2   5162         1726
## 5 22   4136         1635

grap.pop <- aggregate(swiss1[,2:3],by=list(swiss1$CT),sum)
grap.pop[1:5,]

##   Group.1 HApoly Surfacesbois
## 1      1 166093         53088
## 2      2 584102        184554
## 3      3 142915         44879
## 4      4 105751         19626
## 5      5  85151         30589

cor(grap.pop$HApoly,grap.pop$Surfacesbois)

## [1] 0.9601296
# [1] 0.9601296

sum(grap.pop$HApoly)

## [1] 3998831
# [1] 3998831

nI <- 6
pik.swiss <- inclusionprobabilities(grap.pop$HApoly,nI)

poids <- 1/pik.swiss
set.seed(123456)
sys_swiss <- UPsystematic(pik.swiss)
ech.sys_grap <- svydesign(id=~1,weights=poids[sys_swiss==1],
                        data=grap.pop[sys_swiss==1,])

ech.sys_grap$variables

##   Group.1 HApoly Surfacesbois
## 2      2 584102        184554
## 8      8  68069         20673
## 18     18 710547        189739
## 19     19 139490         51785
## 22     22 282264        102248
## 24     24  71695         31204
```

```

pik.swiss[sys_swiss==1]

## [1] 0.8881563 0.1035023 1.0000000 0.2121015 0.4291965 0.1090158
# [1] 0.8881563 0.1607997 0.2966091 1.0000000 0.4162627 0.7926718

nI*ech.sys_grap$variables$HApoly/sum(grap.pop$HApoly)

## [1] 0.8764091 0.1021333 1.0661321 0.2092962 0.4235198 0.1075739
# [1] 0.8764091 0.1586729 0.2926860 1.0661321 0.4107570 0.782187

svytotal(~Surfacesbois,ech.sys_grap)

##              total      SE
## Surfacesbois 1365885 87821
#total SE
#Surfacesbois 1365885 87821

```